1.Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

- **System Used: Windows**

1. **Assign a Static IPv6 Address**

   - Open **Command Prompt as Administrator** and run:

netsh interface ipv6 set address name="Wi-Fi" address=2001:db8:1::10

2. **Verify the Configuration**

   - Run:

ipconfig

The output should show:

IPv6 Address: 2001:db8:1::10

Result: The static IPv6 address **2001:db8:1::10** has been successfully assigned and verified on the Wi-Fi network adapter.

2.Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address. Hint: Use the ping6 or ping -6 command for IPv6 testing.

- **System Used: Windows**

1. **Assign a Unique Local IPv6 Address**
   Open Command Prompt as Administrator and run:

```
netsh interface ipv6 set address name="Wi-Fi" address=fd00:1:1::10
```

2. **Configure IPv6 Address on Another LAN Device**
   Example address for the second device:

```
fd00:1:1::20
```

3. **Test IPv6 Connectivity**
   From the first computer, run:

```
ping -6 fd00:1:1::20
```

The output should show:

```
Reply from fd00:1:1::20: time<1ms
```

Reply from fd00:1:1::20: time<1ms

Result: The unique local IPv6 address fd00:1:1::10 was configured successfully, and connectivity with the LAN device fd00:1:1::20 was verified using ping -6.

3.Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect. Constraint: Document the steps and the observed address behavior in a text file.

- System Used: Windows

1. **Enable Temporary IPv6 Addresses**
   Open Command Prompt as Administrator and run:

```
netsh interface ipv6 set privacy state=enabled
```

2. **Check the Current IPv6 Address**
   Run:

```
Ipconfig
```

Example output:

Temporary IPv6 Address: 2001:db8:1::abcd

3. **Disconnect and Reconnect the Network**
   Disconnect the Wi-Fi, reconnect it, and run:

```
Ipconfig
```

Check the Temporary IPv6 Address again.

4. **Restart the Computer**
   After restarting, run:

Ipconfig

Check whether the temporary IPv6 address has changed.

Result: The temporary IPv6 address was configured and checked before and after network reconnection/reboot. The observed address can change because temporary IPv6 addresses are designed to provide privacy.

4.Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

- **Language Used: Python**

- **Create the Python Script:**

```
restaurants = {
    "Pizza Hut": "2001:db8:1::10",
    "Dominos": "2001:db8:1::20",
    "Burger King": "2001:db8:1::30",
    "Subway": "2001:db8:1::40"
}

search = input("Enter restaurant name: ").lower()

found = False

for name, ipv6 in restaurants.items():
    if search in name.lower():
        print("Restaurant:", name)
        print("IPv6 Address:", ipv6)
```

```
        found = True

if not found:
    print("Restaurant not found.")
```

- **Example Output:**

```
Enter restaurant name: pizza

Restaurant: Pizza Hut
IPv6 Address: 2001:db8:1::10
```

5.Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

- **IPv6 Subnetting Scenario and Manual Configuration**

1. **AI-Generated IPv6 Subnetting Scenario**

Suppose a company has the IPv6 network:

2001:db8:1234::/48

The company needs at least 3 separate subnets. We can use a /64 prefix for each subnet.

Subnet 1: Administration

2001:db8:1234:1::/64

Example host address:
2001:db8:1234:1::10/64

Subnet 2: Development

2001:db8:1234:2::/64

Example host address:
2001:db8:1234:2::10/64

Subnet 3: Testing

2001:db8:1234:3::/64

Example host address:
2001:db8:1234:3::10/64

I selected Subnet 1 (2001:db8:1234:1::/64) for my system.

2. **Configuration Steps on Windows**

1. **Open Command Prompt as Administrator.**
2. **Find the network adapter name:**

netsh interface ipv6 show interfaces

3. **Identify the active network adapter, such as Wi-Fi or Ethernet.**

4. **Assign the IPv6 address using:**

netsh interface ipv6 add address "Wi-Fi" 2001:db8:1234:1::10

Replace Wi-Fi with the name of the active adapter if necessary.

5. **Verify the configuration using:**

Ipconfig

6. **Check that the manually assigned IPv6 address appears under the selected network adapter.**

3. Configured Address

IPv6 Address:
2001:db8:1234:1::10/64

Subnet:
2001:db8:1234:1::/64

4. Result

The IPv6 subnet 2001:db8:1234:1::/64 was selected from the AI-generated scenario, and a valid IPv6 address was manually assigned to the system's network adapter. The configuration was verified using ipconfig.